Stationary profiles of strict Newton trees

Abstract

The Newton evaluation of a finite Conway expression is a sum over integer labels on a tree. We characterize its exact positive support and its strict real states for arbitrary ordered products and turns. After adjacent product vertices are contracted, strictness is equivalent to having at most one turned leaf at each remaining vertex. The original wire lattice is primitive. A direct boundary argument proves that its complete positive action has a unique interior maximum, with negative definite Hessian, also on every admissible fixed-output slice.

We determine the stationary boundary profiles and their additive constants. The profiles of the two one-sided output types differ from the elementary profile only by intrinsic satellite actions; their parameters obey an exact harmonic addition law. At every closed strict maximum the original leaf indices are at least one quarter. These statements retain the original integer support and the complete positive action, providing the variational data for separate source estimates and marked geometric constructions.

1 Introduction and exact inputs

The boundary of a recursively constructed tangle need not behave like the boundary of a completed pair of elementary pieces. A turn can produce an output whose allowable indices occupy only the upper half of the interval. Such a piece can still be attached to a lower or unrestricted output, and the resulting closed expression can have a strict stationary state. This paper follows the actual integer supports through those operations and studies their stationary profiles.

The underlying finite calculus is the Newton calculus of [1]. We use its multiplication, turn, circle and closure identities, which hold in the full polynomial quotient, including the first derivatives at repeated spectral roots. We extend its variational argument from completed/completed roots to all strict ordered expressions. In particular, we give the endpoint descent and angle-map arguments on the enlarged syntax; interior existence is not inferred from a negative Hessian.

The main finite and variational results are Theorems 2.5 and 3.3. Theorems 4.3 and 4.4 determine the absolute boundary profiles. Theorem 5.1 places every closed stationary leaf in $[1/4, 1/2)$. The elementary positive-sum limit in Theorem 3.3 identifies that maximum with the rate of the original positive evaluation. All expressions and their number of leaves are fixed before $N \to \infty$.

The analytic action below is a sum of elementary ideal-tetrahedron volume functions. Its concavity is the familiar angle-variable calculus discussed in [2]; we give the derivatives and the required boundary proof explicitly. The tangle convention is the ordered convention of [1], within the usual framework of Conway composition [3]. No classification up to arbitrary tangle isotopy is needed.

1.1 The original expressions

Each occurrence of h is a distinct turned auxiliary circle around two main strands. In a stack FG, F lies below G. The endpoints, in cyclic order, are bottom left, bottom right, top right

and top left; the exposed tuple after stacking is (F_0, F_1, G_2, G_3), with joins F_2G_1, F_3G_0. The clockwise turn T sends the exposed tuple to (F_1, F_2, F_3, F_0). Cylindrical closure $C(FG)$ joins bottom to top on both sides. The arbitrary open syntax considered here is

$$E ::= h \mid \mathsf{T}E \mid E_1E_2.$$

The original expression retains this order and every written leaf. Algebraic regroupings below compute its source; they do not discard physical circles or assert new isotopies.

Fix $N \geq 2$, and set

$$d = N-1, \quad M = \lfloor d/2 \rfloor, \quad q = e^{\pi \mathrm{i}/N}, \qquad S_r = \prod_{a=1}^{r} 2\sin(\pi a/N), \quad S_0 = 1.$$

Let C_N be the Chebyshev polynomial characterized by $\mathsf{C}_N(\cos t) = \cos Nt$. The full pair algebra is

$$\mathcal{H}_N = \mathbb{C}[z]/(P_N(z)), \qquad P_N(z) = 2\mathsf{C}_N(z/2) + 2.$$

Its normalized Newton basis is

$$e_r(z) = S_r^{-2} \prod_{a=0}^{r-1} (c_a - z), \qquad c_a = 2\cos((2a+1)\pi/N), \quad 0 \leq r \leq d.$$

The nodes c_a repeat, except for one simple node at odd level. Equality in $\mathcal{H}_N$ includes the first derivatives at the double nodes.

Proposition 1.1 (Imported finite calculus)**.** *In this basis the actual turn, marked closure ϕ and multiplication satisfy*

$$\mathsf{T}e_r = e_{d-r}, \qquad \phi(e_ie_j) = \mathbf{1}_{i+j=d}, \tag{1.1}$$

$$e_ie_j = \sum_{r=\max(i,j)}^{i+j} \mu_N(i,j,r)e_r \qquad (i+j \leq d),$$

$$\mu_N(i,j,r) = \frac{S_{i+j}S_r}{S_iS_jS_{r-i}S_{r-j}S_{i+j-r}} > 0, \qquad e_ie_j = 0 \quad (i+j > d). \tag{1.2}$$

The complete pure leaf is

$$h_N = \sum_{r=0}^{M} a_{N,r}e_r, \qquad a_{N,r} = \frac{N^2S_{2r}}{S_r^4} > 0. \tag{1.3}$$

These are [1, Proposition 3.2, Theorem 3.3, Proposition 4.1 and equation (4.3)]. They use the original marked normalization.

For later colored expressions, write $L_j = (-1)^{j-1}U_{j-1}(z/2)$, where U_n is the second-kind Chebyshev polynomial, $U_0 = 1$, $U_1(z/2) = z$ and $U_{n+1}(z/2) = zU_n(z/2) - U_{n-1}(z/2)$. It is the actual circle operator of color j. The actual positive crossing in the pair algebra is

$$R = \kappa \sum_{r=0}^{d} (-q)^r e_r, \qquad \kappa = q^{d^2/2},$$

by [1, equation (4.8)]; its inverse is the negative crossing. Thus an independently colored leaf with retained sign ε means exactly $h_j^{\varepsilon} = \mathsf{T}(R^{\varepsilon}L_j)$. These definitions retain the physical colors and signs in the exact rerooting identities below. The positive source Z_N^0 is the scalar obtained from a fixed $C(FG)$ by these rules with every leaf equal to h_N. Its closing product has the exact coefficient-one relation $i+j=d$. We delete that forced gate only after imposing the relation. All other products remain.

2 Exact supports and reduced product junctions

A support is a set of Newton indices with nonzero coefficients. For the complete pure source these coefficients are nonnegative, so supports can be computed without cancellations.

Proposition 2.1 (Integer supports)**.** *Every nonempty open complete support is an integer interval. If $I_E = [l_E, u_E] \cap \mathbb{Z}$, then*

$$I_h = [0, M] \cap \mathbb{Z}, \qquad I_{\mathsf{T}E} = [d - u_E, d - l_E] \cap \mathbb{Z},$$
$$I_{FG} = \begin{cases} \varnothing, & l_F + l_G > d, \\ [\max(l_F, l_G), \min(d, u_F + u_G)] \cap \mathbb{Z}, & l_F + l_G \le d. \end{cases} \tag{2.1}$$

Every coefficient in the displayed nonempty interval is positive. An empty support remains empty under subsequent operations.

Proof. Only the product requires proof. For a proposed output r, the allowable inputs lie in $[l_F, \min(u_F, r)]$ and $[l_G, \min(u_G, r)]$. Their sums fill an integer interval with lower endpoint $l_F + l_G$ and upper endpoint $\min(u_F, r) + \min(u_G, r) \ge r$. Both r and the lower endpoint are at most d, so this sum interval meets $[r, d]$. Formula (1.2) supplies a positive summand. The same formula gives every necessary inequality and the zero case. □

In particular

$$\phi(FG) = \sum_{r=0}^{d} [e_r]F\,[e_{d-r}]G.$$

This identity also holds for arbitrary signed leaf coefficients. Only the positive-support conclusion uses positivity.

2.1 Strict real supports

Scale the finite wire labels by d. A strict leaf label is $0 < a < 1/2$. A strict retained product with inputs x, y and output z satisfies

$$\max(x, y) < z < x + y < 1. \tag{2.2}$$

A turn sends x to $1 - x$. At a closed root impose $x + y = 1$ instead of requiring a strict final product.

Lemma 2.2 (The complete type table)**.** *The set of strictly realizable outputs of an open expression is exactly one of*

$$L = (0, 1/2), \qquad H = (1/2, 1), \qquad A = (0, 1), \qquad \varnothing.$$

The leaf has type L. Turn exchanges L, H, fixes A, and preserves emptiness. Product is symmetric and has table

	L	H	A
L	A	H	A
H	H	$\varnothing$	H
A	A	H	A

.

The strict closed pairs are AA, AL, AH, LH and their reversals.

Proof. For two low inputs and any $0 < z < 1$, choose both inputs in $(z/2, \min(z, 1/2))$. For an L, A pair choose a sufficiently small low input $c > 0$ and the other input $z - c/2$. Two A inputs can use the first construction. A high input forces $z > 1/2$; for such z choose that input $z - c/2$ and the other input $c > 0$ sufficiently small. Their sum is less than one. Two high inputs cannot have sum less than one. These choices prove sufficiency at every output in each stated interval, as well as necessity. Finally, the closed condition is $S_F \cap (1 - S_G) \neq \varnothing$. □

Definition 2.3. A cylindrical expression is *strict* if it has a strict reduced real state in this sense. This definition concerns its original wire constraints.

An empty strict type does not imply finite vanishing. For example $\mathsf{T}h\,\mathsf{T}h$ has no strict state. At even N its support is empty, whereas at odd $N = 2m + 1$ its support is the full interval $[m, d]$. Both original leaf indices are then forced to their upper endpoint m, so the contribution is boundary-only despite this range of product outputs. Its cylindrical scalar at output d is positive. These assertions follow from (2.1).

2.2 A product with several inputs

The symmetric tensor $C_k(i_1, \ldots, i_k) = \phi(e_{i_1} \cdots e_{i_k})$ retains its normalization from (1.2). For $k = 3$ its support is

$$i_1 + i_2,\ i_1 + i_3,\ i_2 + i_3 \leq d, \qquad i_1 + i_2 + i_3 \geq d.$$

The input legs of a product gate are $x, y, 1 - z$ in this symmetric notation.

Lemma 2.4 (Multiple-input support)**.** *For $n \geq 2$ the product of basis elements with indices $i_1, \ldots, i_n$ is zero unless every pair satisfies $i_p + i_q \leq d$. If all these inequalities hold, its positive support is*

$$[\max_p i_p, \min(d, \sum_p i_p)] \cap \mathbb{Z}. \tag{2.3}$$

For fixed $0 < x_p < 1$, a prescribed output z has a strict resolution in any binary parenthesization precisely when

$$x_p + x_q < 1 \ (p \neq q), \qquad \max_p x_p < z < \min(1, \sum_p x_p). \tag{2.4}$$

A closed k-input product, $k \geq 3$, has a strict resolution precisely when

$$\sum_p x_p > 1, \qquad x_p + x_q < 1 \quad (p \neq q). \tag{2.5}$$

Proof. Associate a violating pair first to prove the finite zero assertion. Otherwise split at any binary root. The lower endpoints of the two child supports are their largest indices; their sum is at most d by the pair inequalities. Proposition 2.1 proves (2.3) inductively, for this and every other parenthesization.

For necessity in the strict statement, follow two inputs to their first common gate. Its child labels are at least those two input labels, and their sum is less than one. Following outputs also gives the strict maximum and sum bounds. For sufficiency, in a child set S write $L_S = \max_{p \in S} x_p$, $U_S = \min(1, \sum_{p \in S} x_p)$. A nonsingleton child has output interval (L_S, U_S); a singleton has its prescribed value. Restrict the two children to values below z. The lower endpoint of their possible sum is $L_1 + L_2 < 1$, and its upper endpoint is greater than z, by (2.4). Thus their sum can be chosen strictly between z and one. If both are singletons, their fixed sum already lies there. This gives a strict parent gate, and induction fills its children.

For a closed product split at the removed final gate. If both child sets have at least two inputs, their lower endpoint sum is less than one and their upper endpoint sum is greater than one, so their output intervals contain a complementary pair. If one set is the singleton x, the other interval contains $1 - x$: its lower bound is smaller by the pair inequalities, and its upper bound is larger by the total-sum inequality. This proves sufficiency. For necessity, two inputs meeting at a retained gate have sum less than one. If their first common vertex is the deleted closing gate, at least one branch contains a strict product because $k \geq 3$; its output strictly exceeds the selected input. The two selected inputs therefore still have sum less than one. The same strict product makes the total of the external inputs strictly larger than the two closing outputs, whose sum is one. For $k = 2$ only that relation remains, namely $x_1 + x_2 = 1$. □

2.3 The reduced junction tree

Write $\mathbf{J}_{ij} = \mathbf{1}_{i+j=d}$. Raising one index of C_3 by $\mathbf{J}$ recovers multiplication. Thus an internal edge without a turn contracts through $\mathbf{J}$, while an edge with a turn contracts through $\mathbf{J}\mathsf{T} = I$. Contract unary chains modulo the algebraic identity $\mathsf{T}^2 = I$, suppress the degree-two closing vertex, and collapse every internal edge without a turn. The result is a tree of product vertices of valence at least three. We call these vertices *junctions*. Every remaining internal edge has one turn. The original leaf labels are retained. A leaf is *flagged* if its incident edge has one turn: it supplies $\mathsf{T}h$ rather than h.

Theorem 2.5 (Reduced junction criterion)**.** *For a closed expression with $H \geq 3$ leaves, the reduced junction tree is unique under the indicated contractions. The expression is strict if and only if each junction has at most one flagged leaf. Its full finite source is the contraction of the same tensors C_k, with all independent leaf coefficients and repeated-root normal data retained.*

Proof. The contracted components are exactly the connected components of the internal edges without a turn, so they are uniquely specified. Frobenius associativity identifies their tensor with C_k. Lemma 2.4 proves preservation of strict feasibility at fixed external labels under every collapse and resolution.

Two flagged leaves at a junction have labels greater than $1/2$, contradicting (2.5). Conversely root at an internal edge, if one exists. Every junction has at least two branches besides its parent. An unflagged leaf supplies type L and a flagged leaf type H. Inductively an outward child product has type H if its own junction has one flagged leaf, and A otherwise. After the intervening turn its input to the parent has type L or A. The table of Lemma 2.2 therefore gives exactly these claimed outgoing types. At the chosen root edge one side is turned, leaving one of the closing pairs AA, AL, HA, HL, each strict. For a single junction, expose a leaf. With no flag the remaining product has type A and the pair is L, A. With one flag expose it, giving H, A. The strict resolution lemma transports the resulting state back to the literal binary syntax. □

With two leaves there is no junction. The strict case has one turn on their connecting edge and the two original leaf labels are equal. We retain this case separately below.

For later estimates it is useful to record the exact outgoing coupons. Root at a flagged leaf when one is present. A nonroot junction gives

$$Q_v = \begin{cases} \mathsf{T}(p_{i(v)} \prod_w Q_w \prod_l h_l), & \text{one flagged leaf,} \\ \mathsf{T}(\prod_w Q_w \prod_l h_l), & \text{no flagged leaf,} \end{cases} \qquad p_i = \mathsf{T}h_i. \tag{2.6}$$

The final scalar is $\phi(p_0 \prod Q_w \prod h_l)$. These are exact full-algebra identities, also for $h_i = \mathsf{T}(R^{\varepsilon_i} L_{j_i})$ at independent colors and signs. The two outgoing types are respectively L and A.

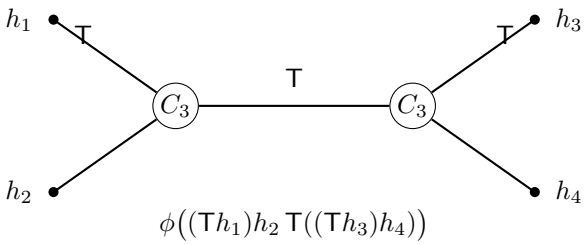


Figure 1: A strict source with four original leaves. Each trivalent vertex carries the symmetric product tensor. The middle edge has a turn, so it is retained. A flagged leaf supplies $\mathsf{T}h_i$; the other leaves supply h_i. Each vertex has one flagged leaf. The labels i remain independent when colors or crossings are changed.

Commutativity and $\mathsf{T}^2 = I$ in this calculation are not claims that arbitrary physical turns or products can be removed from the underlying tangle.

Corollary 2.6 (Exposing any original leaf)**.** *Every original leaf of a strict closed tree with $H \geq 3$ can be exposed as a closing character without changing any other leaf. If its reduced edge parity is σ, and A is the product of the other branches at its junction, then*

$$Z_N = \phi((\mathsf{T}^\sigma h_j)A) = \phi(h_j \mathsf{T}^\sigma A) = \phi(p_j \mathsf{T}^{\sigma+1} A), \qquad p_j = \mathsf{T}h_j.$$

The last factor is either an unturned product of at least two L/A branches, a decorated low module of the first form in (2.6), or the turned output of an unflagged product with at least two branches.

Proof. The pairing in (1.1) makes T self-adjoint: $\phi((\mathsf{T}F)G) = \phi(F(\mathsf{T}G))$. This proves the identities. If $\sigma = 1$, the root leaf uses the only possible flag, so all other branches are L/A and $\mathsf{T}^{\sigma+1}A = A$. If $\sigma = 0$ and another leaf at the junction is flagged, $A = p_i \prod Q$ and $\mathsf{T}A$ is the decorated low module. If there is no such flag, $\mathsf{T}A$ is the outgoing unflagged product. The valence is at least three in all cases. In particular an entirely unflagged three-leaf tree is included. □

For a two-input product, two high Newton coordinates vanish exactly: if $i, j > M$, then $i + j > d$ at both parities. This finite observation is useful when the last case of Corollary 2.6 has only two inputs. It does not by itself bound a signed character at the exposed root.

3 The primitive polytope and its interior maximum

For a fixed strict expression let P be the compact polytope obtained by replacing all strict leaf and product inequalities by their weak versions. Turns retain $y = 1 - x$, and closure retains the complementary relation. Its relative interior is the strict state set: a strict state exists, so none of the listed proper inequalities is identically an equality.

3.1 Integer coordinates and endpoints

Lemma 3.1 (Primitive coordinates)**.** *A closed tree with H leaves has $K = 2H - 3$ free integer coordinates after the forced closing product is removed. Every remaining finite label has the form*

$$r_w = N L_w(k/N) + \delta_w, \qquad k \in \mathbb{Z}^K, \tag{3.1}$$

where L_w is integral affine and δ_w is a fixed integer. There is no residual congruence restriction. Every strict real state is approached by admissible states at all sufficiently large levels of

both parities. For an open tree the full dimension is $2H-1$ *and its fixed-output slice has dimension* $2H-2$.

Proof. Keep a coordinate for every leaf output and every product output. A unary turn simply replaces its input by $d-r$. The closed tree has $H-2$ retained products, in addition to its H leaves. Its two root subtrees are disjoint, so the closing relation can be solved for an output coordinate in one subtree with coefficient 1 or -1. No identification of the two root coordinates has been made. In particular, this procedure never creates $2r=d$. This proves the dimension and primitivity assertions. Replacing d by $N-1$ gives (3.1).

Round d times the free coordinates of a strict state, and reconstruct the eliminated coordinate exactly. The errors in all labels are bounded for the fixed expression. Every proper inequality has a positive slack times d, so the rounded state is admissible for all sufficiently large N. The strict leaf slack also puts its labels below M, at either parity. For a prescribed open output $r_N/d \to b$ in the actual strict interval, choose strict states continuously near b, fix that output first and round the other coordinates. Such a continuous choice is obtained by interpolating two nearby strict states with output on either side of b. □

Define

$$J(t) = \int_0^t \log(2\sin \pi u)\,\mathrm{d}u, \qquad 0 \le t \le 1, \tag{3.2}$$

$$g(a) = J(2a) - 4J(a) = -2J(a) - 2J(1/2-a), \tag{3.3}$$

$$\begin{aligned} m(x,y,z) = {} & J(x+y) + J(z) - J(x) - J(y) \\ & - J(z-x) - J(z-y) - J(x+y-z). \end{aligned} \tag{3.4}$$

Here J has its continuous endpoint values, $J(1-t) = -J(t)$ and $J(1/2) = 0$. The complete action is the sum of one g for each original leaf and one m for each retained product:

$$\mathcal{A}(y) = \sum_{\ell=1}^{H} g(a_\ell(y)) + \sum_{v=1}^{H-2} m(x_v(y), y_v(y), z_v(y)). \tag{3.5}$$

For an open expression the sum contains all its products.

The uniform factorial estimate [1, Lemma 5.2] is $\log S_r = NJ(r/N) + O(\log N)$ for $0 \le r < N$. Applied to (1.2) and (1.3), it gives

$$\log w_N(\mathbf{r}) = N\mathcal{A}(\mathbf{r}/d) + O_E(\log N) \tag{3.6}$$

uniformly on all nonzero positive states of a fixed expression. In using this estimate at the endpoints, the modulus of continuity of J is $O(\eta|\log\eta|)$. Thus the changes from r/N to r/d, and from M/N to $1/2$, contribute $O_E(\log N)$, even at the support boundary.

3.2 The atoms and their boundary derivative

For $a+b+c=1$ set

$$V(a,b,c) = -\pi\{J(a) + J(b) + J(c)\}.$$

This is a nonnegative function, zero on the boundary of the angle simplex and positive in its interior. Only these elementary properties and its derivatives are used here. With $c = 1-a-b$ its Hessian has entries

$$-\pi^2 \begin{pmatrix} \cot \pi a + \cot \pi c & \cot \pi c \\ \cot \pi c & \cot \pi b + \cot \pi c \end{pmatrix}.$$

Both displayed diagonal sums are positive, and the determinant is π^4, since $\cot \pi a \cot \pi b + \cot \pi b \cot \pi c + \cot \pi c \cot \pi a = 1$. The Hessian is therefore negative definite at a positive triple.

At a product put

$$p = z - x, \quad q_0 = z - y, \quad r = x + y - z, \quad w = 1 - x - y.$$

The subscript on q_0 distinguishes this angle from the root of unity. Direct substitution gives

$$\pi g(a) = 2V(a, 1/2 - a, 1/2), \tag{3.7}$$

$$\pi m(x, y, z) = V(y, x, w) + V(p, q_0, 2r + w) + V(p + q_0, r, r + w). \tag{3.8}$$

All three entries of each atom are nonnegative and sum to one on the original gate support. Consequently the whole action is concave on its shared-label polytope.

There are two different kinds of flat atom. At $(0, a, 1 - a)$ with $0 < a < 1$, the directional derivative toward a positive atom is $+\infty$: $-J(t) = -t \log t + O(t)$ near zero. At $(0, 0, 1)$ the singularities cancel. More explicitly, along the segment with small entries $t\alpha, t\beta$,

$$V(t\alpha, t\beta, 1 - t(\alpha + \beta)) = \pi t\{(\alpha + \beta) \log(\alpha + \beta) - \alpha \log \alpha - \beta \log \beta\} + O(t^2).$$

The derivative is finite. A positive atom also has a finite derivative. Thus no atom contributes $-\infty$ along a segment toward a strict state.

Lemma 3.2 (Endpoint descent)**.** *At every nonstrict point of a nonempty strict closed polytope, or of an open fixed-output slice with output in its strict interval, at least one atom in (3.7)–(3.8) is flat but is not a vertex of its angle simplex.*

Proof. First suppose a proper wire has value zero or one. A turn replaces it by its complementary endpoint. At a product with output zero both inputs are zero. At a product with output one the inputs sum to one: if both are interior, the first atom is $(y, x, 0)$, as required; otherwise one may descend to an endpoint child. Repeating this descent in the finite tree terminates at a leaf endpoint. A leaf of value zero has atom $(0, 1/2, 1/2)$; a leaf cannot have value one. The same argument applies to either child of the deleted root: one descends inside that child, without putting the removed flat gate back.

Suppose now that every wire is in $(0, 1)$. A leaf at its allowed upper endpoint $1/2$ gives $(1/2, 0, 1/2)$. If every leaf is strict, a nonstrict state must lie on a retained gate face. When $w = 0$, the first atom $(y, x, 0)$ is nonvertex-flat. When $r = 0$, the third atom is $(z, 0, 1 - z)$. When $p = 0$ and $q_0 > 0$, the second atom has one zero entry and two positive entries; the case $q_0 = 0 < p$ is symmetric. In the remaining intersection $p = q_0 = 0$, one has $x = y = z \in (0, 1)$ and the third atom is $(0, x, 1 - x)$. This lists every gate boundary.

For a fixed-output slice the output itself is in $(0, 1)$, and the same descent applies to each proper endpoint wire. The segment used below keeps that output fixed. No boundary caused only by imposing the output is included among these proper faces. □

Theorem 3.3 (Unique strict optimizer)**.** *Every strict closed expression has a unique maximizer y_* of its complete action. It is strict and its Hessian in the primitive coordinates is negative definite. The positive scalar satisfies the all-level limit*

$$\lim_{N \to \infty} \frac{1}{N} \log Z_N^0 = F_0, \qquad F_0 = \mathcal{A}(y_*) > 0. \tag{3.9}$$

For an open expression of nonempty type, each output b in its strict interval has a unique strict fixed-output maximizer. The profile

$$f_E(b) = \max_{\text{output } b} \mathcal{A}_E$$

is real analytic and strictly concave on that interval. It has a continuous extension to the closed feasible output interval.

Proof. Compactness gives a maximizer. Join a nonstrict point to a strict state in the same polytope. By Lemma 3.2, one atom contributes $+\infty$ to the one-sided derivative, and none contributes $-\infty$. Hence the action increases and the point cannot maximize.

At a strict product the gate Hessian is already negative definite in (x, y, z). Indeed the first atom in (3.8) detects $(\delta x, \delta y)$; if those vanish, the second detects δz. Its third atom is nonpositive and cannot cancel either negative term. The leaf Hessian is strictly negative since

$$g'(a) = 2 \log \cot \pi a, \qquad g''(a) = -\frac{4\pi}{\sin 2\pi a} < 0.$$

If all atom variations vanish, every leaf output and retained product output is fixed. Turns then fix their output wires, and the closing relation fixes the eliminated coordinate. Thus the sum is negative definite on the primitive tangent space. This proves uniqueness by concavity.

For an open expression the same argument applies including its output coordinate: its last product detects that coordinate, or the expression is a turned leaf and its leaf detects it. On a fixed-output slice, the internal Hessian is therefore invertible. The analytic implicit-function theorem gives a real analytic maximizing state and profile. The full negative Hessian has a negative Schur complement in the output coordinate, so $f_E'' < 0$. For the single-leaf case this is the displayed formula and its reflection.

Continuity at a feasible endpoint is worth recording. For any endpoint maximizing state, interpolate it with an interior strict state. This supplies states at every nearby interior output, and their actions converge to the endpoint maximum. Conversely, every sequence of maximizing states has a convergent subsequence in the compact full polytope; its limit is feasible at the endpoint. These two observations prove continuity. They do not require finite support at the exact endpoint for every parity.

There are at most N^K positive states. Equation (3.6) gives the upper limit in (3.9). Primitive rounding of y_* gives a genuine positive summand at every sufficiently large N and the matching lower limit. The strict leaf atoms are positive, so $F_0 > 0$. □

Lemma 3.4 (Gate curvature without leaf curvature)**.** *For a closed strict tree with $H \geq 3$, the sum of the retained gate Hessians is itself negative definite in the primitive coordinates.*

Proof. Contract unary turn chains. Every wire in a nontrivial root subtree appears at a retained product, either as an input or as an output. If all gate variations vanish, these wires have zero variation. The only possible uncovered root subtree is a single leaf with turns; its variation is fixed by the complementary root relation and the other subtree. Both subtrees can be single leaves only when $H = 2$. The injectivity and the strict gate Hessian proved above give the result. □

4 Stationary boundary profiles and their constants

The derivative of a fixed-output profile describes how the maximum changes when the output moves. It is convenient to record this derivative together with the output itself. For a strict open output b, define

$$B = \cot \pi b, \qquad \kappa_E = e^{-f_E'(b)}, \qquad X = \sqrt{\frac{1 + B^2}{1 + \kappa_E}}, \qquad Y = \sqrt{\kappa_E}\, X. \tag{4.1}$$

The square roots are positive. Thus $X^2 + Y^2 - B^2 = 1$ and $f_E'(b) = 2 \log(X/Y)$. We call (B, X, Y) the stationary boundary parameters. Their definitions are variational; no global geometric realization is assumed.

Lemma 4.1 (The stationary product)**.** *At a strict stationary product the two children and the output have the same X. With child cotangents B_1, B_2,*

$$S = B_1 + B_2 > 0, \qquad B_o = \frac{B_1 B_2 + X^2 - 1}{S}, \qquad Y_o = \frac{Y_1 Y_2}{S}. \tag{4.2}$$

A turn sends (B, X, Y) to $(-B, Y, X)$.

Proof. At inputs x, y and output z, stationarity gives $f_1'(x) = -m_x$, $f_2'(y) = -m_y$ and $f_o'(z) = m_z$. Differentiating (3.4) yields

$$\begin{aligned} e^{m_x} &= \frac{\sin\pi(x+y)\sin\pi(z-x)}{\sin\pi x \sin\pi(x+y-z)}, \\ e^{m_y} &= \frac{\sin\pi(x+y)\sin\pi(z-y)}{\sin\pi y \sin\pi(x+y-z)}, \\ e^{-m_z} &= \frac{\sin\pi(z-x)\sin\pi(z-y)}{\sin\pi z \sin\pi(x+y-z)}. \end{aligned}$$

Set $D = 1 - B_1 B_2 + B_o(B_1 + B_2)$. The identities

$$S = \frac{\sin\pi(x+y)}{\sin\pi x \sin\pi y} > 0, \qquad D = \frac{\sin\pi(x+y-z)}{\sin\pi x \sin\pi y \sin\pi z} > 0$$

and (4.1) give

$$X_1^2 = X_2^2 = X_o^2 = D, \quad Y_1^2 = S(B_1 - B_o), \quad Y_2^2 = S(B_2 - B_o), \quad Y_o^2 = (B_1 - B_o)(B_2 - B_o).$$

Taking positive roots proves the product formulas. For a turn $f_{\mathsf{T}E}(b) = f_E(1-b)$, so its derivative changes sign and its cotangent changes sign. This proves the last assertion. □

Proposition 4.2 (Parameters of the three types)**.** *On their actual strict output intervals the parameters satisfy*

$$\begin{array}{c|c} L & (B,X,Y) = (B,B,1), \quad B > 0, \\ H & (B,X,Y) = (-Y,1,Y), \quad Y > 0, \\ A & -Y < B < X. \end{array} \tag{4.3}$$

Consequently there are constants c_E such that

$$f_E(b) = g(b) + c_E \quad (E:L), \qquad f_E(b) = g(1-b) + c_E \quad (E:H). \tag{4.4}$$

Proof. For a leaf, $g' = 2\log\cot\pi b$ gives $(B, B, 1)$. The turn formula exchanges the first two rows and preserves the inequality in the third. Consider the products giving type A, whose children have type L or A. If the common $X < 1$, put $h_0 = \sqrt{1 - X^2}$ and write

$$B_i = h_0 \cot\alpha_i, \qquad Y_i = h_0 \csc\alpha_i, \qquad 0 < \alpha_i < \pi.$$

The low type has $\alpha_i = \arccos X$ and the unrestricted type has $\alpha_i > \arccos X$. The positivity of S gives $\alpha_1 + \alpha_2 < \pi$, and (4.2) adds these angles. Therefore $B_o < X$ and $B_o + Y_o = h_0 \cot((\alpha_1 + \alpha_2)/2) > 0$. If $X \geq 1$, each child has $\sqrt{X^2 - 1} < B_i \leq X$, with equality at the upper bound allowed for type L. In particular $B_o > 0$, and

$$(X - B_1)(X - B_2) < 1$$

implies $B_o < X$ by (4.2). The strict inequality holds also at $X = 1$, since each $B_i > 0$.

A product giving type H has one high child, with parameters $(-u, 1, u)$. Hence the common $X = 1$. Its other child, of type L or A, has parameters $(b, 1, b)$ with $b > 0$. Strictness says $b - u > 0$, and the output is

$$\left(-\frac{ub}{b-u}, 1, \frac{ub}{b-u}\right).$$

This proves all cases of the table. Finally the first two rows determine f'_E as the derivative of $g(b)$ or $g(1-b)$ on the connected strict interval, giving (4.4). Their constants are computed next, without a choice of additive normalization. □

4.1 The intrinsic satellite action

No product has type L. A product of type H has exactly one high child. Starting at an L or H output, follow turns and the unique high child of each product. This path terminates at a unique original leaf, the *distinguished leaf*. The other children on the path are called its satellites, and have type L or A. Between successive high products the number of turns is even. Thus the full algebra also gives

$$E_H = (\mathsf{T}h_*)\prod_{j=1}^{m} Q_j, \qquad E_L = \mathsf{T}\left((\mathsf{T}h_*)\prod_{j=1}^{m} Q_j\right), \tag{4.5}$$

after associating the inherited products. This identity retains the actual polynomials; its stationary profile is simpler than the polynomial.

For a satellite Q define

$$\Omega_Q = \max_{0\le a\le 1/2}\{g(a) + f_Q(a)\}. \tag{4.6}$$

The maximum is the complete action of the literal strict closure $C(h\,\mathsf{T}Q)$. The strict type of Q is L or A, so Theorem 3.3 gives a unique strict maximizer β_Q. Its equation is $g'(\beta_Q) + f'_Q(\beta_Q) = 0$. Consequently the satellite has parameters

$$(B_Q, X_Q, Y_Q) = (b_Q, 1, b_Q), \qquad b_Q = \cot \pi\beta_Q > 0. \tag{4.7}$$

For type L, $\beta_Q = 1/4$ and $b_Q = 1$; for type A, the strict inequality $B_Q < X_Q$ gives $1/4 < \beta_Q < 1/2$ and $0 < b_Q < 1$.

Theorem 4.3 (Harmonic response and absolute profiles)**.** *For the spine (4.5), let u_0 be the cotangent of its distinguished original leaf. Let v be the final high output's Y, equivalently the final low output's X. At its stationary state,*

$$\frac{1}{u_0} = \frac{1}{v} + \sum_{j=1}^{m}\frac{1}{b_{Q_j}}, \qquad c_E = \sum_{j=1}^{m}\Omega_{Q_j}. \tag{4.8}$$

In particular (4.4) holds with exactly this constant. The parameters b_{Q_j} depend only on the corresponding satellites.

Proof. Every high product has common $X = 1$, so its satellite is at the unique state (4.7). With high input $(-u, 1, u)$ and output $(-v, 1, v)$, (4.2) gives

$$v = \frac{ub}{b-u}, \qquad 0 < u < b, \qquad \frac{1}{u} = \frac{1}{v} + \frac{1}{b}. \tag{4.9}$$

Iteration proves the first formula. Conversely every $v, b > 0$ gives the strict branch $0 < u < b$; no sign has been lost by squaring the stationary equations.

It remains to fix the constant. Suppose the incoming high profile is $g(1-x)+c$. Send the outgoing high label z to one. Then $v=-\cot \pi z \to +\infty$, $u \to b$ and $x \to 1-\beta_Q$. The last gate approaches $x+\beta_Q = z = 1$. Its continuous action is zero: in (3.4) use $J(1-\beta_Q) = -J(\beta_Q)$, or note that all three atoms in (3.8) are flat. Continuity of profiles, proved in Theorem 3.3, gives

$$c_{\mathrm{out}} = c + g(\beta_Q) + f_Q(\beta_Q) = c + \Omega_Q.$$

Turns preserve the constant and the bare leaf has constant zero. This proves the second formula. □

For a low satellite, $\Omega_Q = 2G_0 + c_Q$ where $G_0 = g(1/4)$. The formula is an identity of optimized actions. For example at $N=3$ the complete source has $h = 10-z$ and $\mathsf{T}h = z^2-1$; their product is $10z^2-2z-8$ in $\mathbb{C}[z]/((z-1)^2(z+2))$. It is not a scalar multiple of $\mathsf{T}h$: the value ratio at -2 is 12, whereas the normal ratio at 1 is 9. Thus the constant profile does not suppress normal data of the source.

4.2 Unrestricted components and their effective range

Replace each maximal low subtree by its profile $g + c_i$. A remaining type-A subtree has syntax

$$A ::= LL \mid \mathsf{T}A \mid AA \mid LA \mid AL, \tag{4.10}$$

where L here is an effective profile, not a replacement quantum leaf. The definition of the maximum immediately gives

$$f_A(z) = f_{A^{\mathrm{skel}}}(z) + \sum_i c_i. \tag{4.11}$$

At an unflagged junction directed toward a parent, the outgoing expression is $Q = \mathsf{T}A$ with $A = FG$ an unturned product and both children of type L or A. Its intrinsic action is

$$\Omega_Q = \max_{0 \le a \le 1/2} \{g(a) + f_A(1-a)\}.$$

This is the action of the strict closure $C(hA)$.

Theorem 4.4 (The intrinsic probe range)**.** *At this closing maximum the unturned output has*

$$(B_A, X_A, Y_A) = (-x, x, 1), \qquad \frac{1}{2} \le x < 1. \tag{4.12}$$

Every effective leaf in (4.10), including the closing probe, has cotangent in $[1/2, 1)$. Equality at $1/2$ occurs only when the entire effective closure consists of three leaves. In every other case all these cotangents are strictly between $1/2$ and one.

The absolute identities are

$$\Omega_Q = F_{\mathrm{probe}}(A^{\mathrm{skel}}) + \sum_i c_i, \qquad F_0 = \sum_j \Omega_{Q_j} \tag{4.13}$$

for a whole reduced tree rooted at a flagged leaf, with satellites Q_j at that root.

Proof. The root condition says that the two output labels are complementary and their profile derivatives agree. Hence their triples are (B, X, Y) and $(-B, X, Y)$. The bare probe has $(x, x, 1)$, so (4.12) holds initially with $0 < x < 1$, the strict upper bound coming from $-Y_A < B_A$. At the last product use the identities in the proof of Lemma 4.1:

$$1 = Y_A^2 = (B_F - B_A)(B_G - B_A) \le (X_A - B_A)^2 = 4x^2.$$

Both factors are positive because output labels strictly exceed the inputs. This proves $x \geq 1/2$.

At the root product every child has common $X = x < 1$. Its cotangent satisfies $B_i \leq X$ and, by $B_1 + B_2 > 0$, also $B_i > -X$. If it is type A, both inequalities are strict, so $Y_i^2 = 1 + B_i^2 - X^2 < 1$. Turns exchange X, Y. The same argument propagates the strict bounds $X, Y < 1$ through every proper type-A descendant. For a low child $(X, X, 1)$ and another child (B, X, Y), the output bound $Y_o \leq 1$ implies

$$1 + B^2 - X^2 \leq (B + X)^2, \qquad 1 \leq 2X(B + X) \leq 4X^2. \tag{4.14}$$

The last inequality is strict for an unrestricted neighbor. For two low children at a proper descendant, the first is strict because $Y_o < 1$. Thus equality is possible only at the root product with two effective low children. It gives the three-leaf closure and $x = 1/2$ at all three leaves. This proves the range.

The first identity in (4.13) is (4.11) inside the same maximum. For the second, write the original root source as $\phi((\mathsf{T}h_0)\prod Q_j)$. The associated open high profile is $g(1-z) + \sum_j \Omega_{Q_j}$ by Theorem 4.3. At the closing output $z = 1$ the deleted last gate has coefficient one and $g(0) = 0$. Profile continuity identifies its endpoint maximum with precisely the original closed action, proving the formula. □

One boundary case in later coefficient estimates can be read directly from these parameters. If the last product is $A = LF$ with F unrestricted, then

$$B_F = \frac{1}{2x} - x, \qquad Y_F = \frac{1}{2x}, \qquad f'_F = 2\log(2x^2), \quad f'_A = 2\log x.$$

For $1/2 \leq x \leq 2/3$ both slopes lie in $[-2\log 2, 0]$. This is an exact closed parameter range; it does not assert an estimate for the signed source. Such estimates require its actual coefficient arrays.

Corollary 4.5 (The action budget at any exposed leaf)**.** *Use Corollary 2.6, with root edge parity σ and remaining product A. If $\sigma = 1$, every branch of A has type L or A and the original complete action is $F_0 = \sum_Q \Omega_Q$, summed over those branches. If $\sigma = 0$, then $Q = \mathsf{T}A$ has type L or A and $\Omega_Q = F_0$.*

Proof. For $\sigma = 1$, this is the last identity of (4.13) at the exposed flagged root. For $\sigma = 0$, the original positive closure is $C(hA)$, whose action is exactly $\max_a\{g(a) + f_A(1-a)\} = \Omega_{\mathsf{T}A}$. This includes the decorated case in which A has one high child and $\mathsf{T}A$ is low. Only the finite source is rerooted; the variational maximum and every physical leaf are unchanged. □

5 The quarter bound at the original closed maximum

Theorem 5.1 (All original leaves are at or above quarter)**.** *At the unique complete maximum of any strict closed expression, every stationary port satisfies $X, Y \leq 1$. Every original leaf label therefore satisfies*

$$\frac{1}{4} \leq a_\ell(y_*) < \frac{1}{2}. \tag{5.1}$$

Along an L/H spine with m satellites, its distinguished original leaf additionally satisfies $\cot(\pi a_*) \leq 1/(m+1)$.

Proof. At the complementary root the two triples have opposite B and equal X, Y. For type AA the inequalities give $|B| < \min(X, Y)$. Together with $X^2 + Y^2 - B^2 = 1$ this gives $X, Y < 1$. For type AL, the low side is $(x, x, 1)$ and the unrestricted inequality $-1 < -x$

gives $x < 1$. For type AH, the high side has $X = 1$, and the unrestricted side gives $Y < 1$. For type LH, the two explicit forms force $X = Y = 1$. These exhaust the strict closing pairs.

Descend through a product. The common X is already at most one. Every child type satisfies $B_i \le X$, and $B_1 + B_2 > 0$ implies $B_i > -B_j \ge -X$. Thus $Y_i^2 = 1 + B_i^2 - X^2 \le 1$. A turn exchanges the same two bounds. At a leaf, $X = \cot \pi a > 0$, proving (5.1). Finally, in (4.8) one has $v \le 1$ and $b_{Q_j} \le 1$, so $1/u_0 \ge m + 1$. □

The lower endpoint is attained. In $C(h\,\mathsf{T}(h^s))$, the inner s original leaves have $X = 1$ and hence label $1/4$. Indeed their product has at $X = 1$ the parameters $B = Y = 1/s$, its turn has $(-1/s, 1/s, 1)$, and the closing leaf has cotangent $1/s$. Thus strictness of every angle state does not force its original leaf labels to lie strictly above one quarter.

6 Finite reproducibility and subsequent uses

The paper-local program `check.py` uses only the Python standard library. It compares multiplication in the full polynomial quotient with the stated Newton formula in finite fields, retaining the repeated roots. It separately enumerates integer supports, checks the multi-input support theorem under different parenthesizations, and verifies rational instances of the local port and harmonic identities. The output states the finite ranges. These checks detect coefficient and indexing errors; the all-level support, boundary, curvature and positive-sum limit proofs are the arguments above.

The exported variational statements require no global geometric construction. A marked geometric application must assemble the original cells and peripheral curves. A signed-source application must separately establish the local coefficient asymptotics and control the complementary contributions. A full-invariant application must also perform the original filling transform and prove noncancellation. The constants Ω_Q are intrinsic optimized actions; replacing a profile by an effective leaf in Section 4 does not replace its quantum polynomial or its independent physical colors.

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